Monepha Mitchell

Professor Sridhar Ganti

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Advantages of using AI in the Manufacturing Industry

Introduction

Artificial intelligence is becoming important in manufacturing because factories must produce goods quickly, maintain quality, control costs, and respond to changing demand. Traditional automation follows fixed instructions, while AI can analyze production data, identify patterns, and make predictions. Common applications include predictive maintenance, computer-vision inspection, intelligent robotics, production scheduling, and energy management. The National Institute of Standards and Technology explains that machine learning can use sensor data to predict equipment failures, identify defects, and support better production decisions (National Institute of Standards and Technology [NIST], n.d.).

This report examines Siemens Electronics Works Amberg in Germany. The plant shows how AI, industrial sensors, edge computing, cloud analytics, and digital twins can work together to improve manufacturing. The report analyzes the problems addressed, the technologies used, the results achieved, and the implementation risks. It also proposes an AI-based system for reducing scrap and energy use.

Case Study Analysis: Siemens Electronics Works Amberg

Company and Manufacturing Challenge

Siemens Electronics Works Amberg manufactures SIMATIC programmable logic controllers and related electronics. The factory produces approximately 17 million products each year and handles more than 1,000 product variations (Siemens, n.d.). This high-volume, high-mix environment requires the company to maintain quality, prevent machine failures, and avoid production bottlenecks.

One bottleneck occurred during the automatic X-ray inspection of printed circuit boards. Tiny solder joints had to be inspected for correct connections. Siemens could have purchased another X-ray machine for approximately €500,000, but this would have increased costs without improving the intelligence of the production process. Another problem involved a milling spindle used to separate circuit boards. Milling dust contaminated the spindle, leading to unexpected failures and production interruptions (Siemens, n.d.).

AI Technologies and Tools Used

Siemens created a connected system using sensors, edge devices, cloud analytics, machine-learning algorithms, and digital twins. Edge computing allowed data to be processed near the equipment, where production decisions could be made quickly.

For the X-ray process, Siemens trained an AI model using production parameters and previous inspection results. The model learned the relationship between process data and solder-joint quality. It then predicted whether a circuit board was likely to be defect-free and whether a complete end-of-line inspection was necessary. The system did not eliminate inspection. Instead, it helped Siemens focus its inspection resources on circuit boards that were more likely to require additional testing.

For predictive maintenance, Siemens analyzed the milling spindle's rotational speed and electrical current use. A trained model identified abnormal patterns connected with possible failure. Operators could receive warnings approximately 12 to 36 hours before a potential breakdown, allowing maintenance to be scheduled at a less disruptive time (Siemens, n.d.).

Siemens also used digital twins, which are virtual versions of physical production systems, to test equipment designs and production changes. This helped the company improve the planned production cycle time from 11 seconds to its target of eight seconds.

Table 1

Summary of the Siemens Amberg AI Case

Manufacturing challenge	AI or digital tool	Manufacturing benefit
X-ray inspection bottleneck	Machine learning and edge computing	Predicts when additional inspection is needed
Unexpected spindle failures	Predictive-maintenance model	Gives 12–36 hours of advance warning
Production-line design	Digital twin and simulation	Helps reach an eight-second cycle time
High product variety	AI process controls and connected automation	Supports flexible, high-volume production

Outcomes and Benefits

The Siemens case demonstrates several manufacturing advantages. Predictive maintenance changes maintenance from a reactive activity to a planned one. Workers can respond to warning signs before a failure creates downtime, emergency repair costs, damaged materials, or late deliveries. AI-supported quality inspection also allows the factory to use testing equipment more efficiently instead of solving every bottleneck by purchasing another machine.

Digital twins allow engineers to test changes without immediately interrupting physical production. Simulations can reveal bottlenecks, inappropriate equipment, and possible cycle-time problems before expensive purchases are made. Connected AI systems also create a feedback loop in which production results can be used to improve future predictions.

The World Economic Forum reported that the Amberg factory used smart robotics, AI-powered process controls, predictive maintenance, and a lean-digital approach to reach 140 percent of its previous output while handling twice the product complexity, without increasing electricity use or other resources (World Economic Forum, 2021).

However, these results should not be credited to AI alone. They came from a combination of lean manufacturing, automation, data infrastructure, robotics, digital twins, employee knowledge, and AI. This distinction is important because purchasing an AI tool without improving data practices and manufacturing processes may not produce the same results.

Challenges and Risks

AI implementation creates several important risks. Sensor data may be incomplete, inconsistent, or inaccurate. A model trained on poor-quality data may issue false alarms or fail to detect a real problem. Rare failures also create imbalanced datasets because normal operation is much more common than serious defects. A model may appear accurate while still performing poorly on the failures that matter most.

Research on AI applications in quality control and predictive maintenance identifies data imbalance, industry-specific data, model explainability, and real-world performance as continuing challenges (Johanesa et al., 2024). Complex deep-learning systems may also function as “black boxes,” making it difficult for technicians to understand the reasons behind their predictions.

Model drift is another concern. Equipment ages, suppliers change materials, and production settings are adjusted. A model may become less reliable if it is not continuously monitored and periodically retrained. Explainability is especially important because maintenance technicians need to understand why an alert was issued before deciding to stop or service an important machine.

Cybersecurity risks increase when sensors, edge devices, cloud platforms, and production equipment are connected. Manufacturers need access controls, network separation, encryption, software updates, secure backups, and incident-response plans. Other barriers include high implementation costs, legacy-system integration, workforce skill gaps, and employee concerns about job replacement. NIST identifies data availability, cost, workforce readiness, privacy, cybersecurity, and legacy integration as major barriers to adopting AI in manufacturing (NIST, n.d.).

AI should therefore support workers by providing better information while keeping people responsible for important maintenance and quality decisions. Employees should receive training on how the system works, how to interpret its alerts, and how to report inaccurate recommendations.

Proposal for Innovation: AI Resource and Waste Optimization System

A useful new application would be an AI Resource and Waste Optimization System for high-mix manufacturing factories. The system would combine computer vision, machine sensors, energy meters, production records, and a digital twin. Its primary goal would be to reduce scrap and lower the amount of energy used for every acceptable product.

Many factories measure total electricity use and total scrap, but cannot easily identify which machine setting, material batch, work shift, or production step caused the waste. The proposed system would create a detailed digital record for each production batch. Cameras would identify visible defects, while sensors would record temperature, vibration, pressure, speed, power consumption, and cycle time.

A machine-learning model would search for operating conditions connected to product defects or unusually high energy use. A digital twin would then test possible adjustments before they were applied to physical equipment.

For example, the system might discover that a certain temperature and machine-speed combination results in a shortening of the cycle time but increases product defects and electricity use. Instead of optimizing only speed, the AI would consider several goals, including product quality, energy per acceptable unit, production time, and equipment health. It would recommend the settings that provide the best overall result.

Low-risk adjustments could eventually be automated within approved operating limits. Major production changes would continue to require supervisor or engineering approval.

Implementation Plan

The manufacturer should begin with a controlled pilot on one production line. During the first phase, the company would establish baseline measurements for scrap, first-pass yield, energy per good unit, cycle time, and downtime.

During the second phase, sensors, cameras, energy meters, and production records would be connected through a secure data platform. Data scientists and process engineers would then clean the data, identify important variables, and train the AI models. The training data should include defective products and unusual operating conditions rather than only examples of normal production.

The company should test the system’s recommendations through a digital twin or controlled testing environment. NIST emphasizes that industrial AI systems, safeguards, and unexpected behaviors should be evaluated before the systems are introduced into full-scale production (Sharp, 2024).

After successful testing, the system could begin by providing recommendations to operators. Automatic control should be introduced only after the company confirms that the model performs reliably and that safe operating limits are enforced.

Table 2

Proposed Performance Measures

Performance measure	Purpose	Example pilot goal
Scrap rate	Tracks discarded material	Reduce by 10–15 percent
First-pass yield	Tracks products passing without rework	Improve by 5 percent
Energy per good unit	Connects energy use to acceptable output	Reduce by 8–12 percent
False-negative defect rate	Tracks defects missed by the system	Keep below an approved safety limit
Unplanned downtime	Tracks production interruptions	Reduce by 10 percent
Operator acceptance	Measures trust and usability	At least 80 percent positive feedback

Unplanned downtime, Tracks production interruptions, Reduce by 10 percent

Operator acceptance Measures trust and usability, with at least 80 percent positive feedback

The percentages in Table 2 are proposed pilot goals rather than guaranteed outcomes. Actual targets should be based on the factory's baseline performance, budget, equipment condition, and production requirements.

Potential Benefits

The proposed system could lower material and electricity costs, reduce rework, improve first-pass yield, and support sustainability reporting. NIST states that AI has the potential to improve manufacturing productivity, quality control, predictive maintenance, waste reduction, and energy management (NIST, 2025).

The system could also improve root-cause analysis by identifying relationships across several production stages instead of examining each machine separately. Several small variations may combine to create a final defect even when each production step remains within its acceptable limits. An AI system could examine these interactions and provide engineers with evidence showing which variables contributed most to a problem.

Another benefit would be improved decision-making. Managers could compare energy use, quality, production speed, and equipment health through one dashboard. This would help the company avoid reducing energy use in a way that harms quality or increasing production speed in a way that damages equipment.

Anticipated Challenges

The proposal would require an initial investment in sensors, cameras, software integration, cybersecurity, data storage, and employee training. Older machines may not produce information in a usable digital format. The company may need additional devices or custom software to collect data from these machines.

Product images would also need accurate labels, and rare defects could limit model training. Energy goals might conflict with production goals, so managers would need to establish clear priorities and safety limits. The company would also need to monitor the system for model drift as equipment, materials, and production conditions change.

Employees may distrust the system or fear that it will be used to monitor their individual performance. Management should clearly explain that the purpose is to improve the production process rather than to conduct employee surveillance. Workers should participate in system design and provide feedback about alerts and recommendations.

The manufacturer should maintain human review, regular model audits, cybersecurity testing, and a safe manual method for operating when the AI system is unavailable. Final responsibility for safety, maintenance, and product quality should remain with qualified employees.

Conclusion

The Siemens Electronics Works Amberg case shows that AI can create meaningful manufacturing advantages when it is combined with reliable data, connected equipment, edge computing, digital twins, cybersecurity, and employee expertise. Siemens used AI to improve quality inspection, predict milling-spindle failures, and support faster production decisions. Its broader digital transformation increased output and helped the factory manage greater product complexity without a matching increase in resources.

At the same time, AI is not a stand-alone solution. Its value depends on data quality, system integration, workforce readiness, continuous monitoring, and human oversight.

Manufacturers must also address cybersecurity, model drift, inaccurate predictions, implementation costs, and employee trust.

The proposed AI Resource and Waste Optimization System applies to the lessons from Siemens by connecting quality, energy, maintenance, and production information. A carefully controlled pilot could help a manufacturer reduce scrap, lower energy use, improve product quality, and become more resilient. The most responsible approach is to treat AI as a decision-support partner for skilled workers rather than as a replacement for human knowledge and judgment.

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